

07b · The same campaign question, on real data — Google’s geo experiment (companion)

July 12, 2026

The business decision. A real advertiser spent **\$50,000** on a real ad campaign, in **50 US geos, for four weeks** (16 Feb – 15 Mar 2015), and left 50 geos dark. This is the worked example published with Google’s geo-experiment methodology (Kerman, Wang & Vaver 2017, the `matched_markets` library) — daily **sales and ad cost per geo**, exactly the panel your own company’s data warehouse holds. The question is the one notebook 07 trained us for: **how much incremental sales did the spend cause, and did it pay for itself?**

0.0.1 What changes when the data is real

Notebook 07 planted a +12% lift in a simulator and graded synthetic control against it. Here nobody tells us the truth — that is the whole epistemic situation of applied causal inference. Three things replace the planted truth:

1. **A randomized benchmark.** The 100 geos were split into treatment and control **at random** (in matched pairs, as we’ll verify from the data itself). So a design-based estimate — the matched-pairs difference-in-differences — is *unbiased by construction*. It is noisy, but it is the closest real data ever gets to a known answer, and we grade the synthetic control against it.
2. **Falsification tests** that need no truth: placebo campaigns on untreated geos (Depth B), a fake launch date, and a check the simulator could never give us for free — the campaign *ends*, so the estimated effect must **switch off when the money does**.
3. **Real messiness:** missing geo-days, heavy-tailed market sizes (the largest geo sells 300x the smallest), and a weekly cycle you can see with the naked eye. Each one bites somewhere below, and one of them (market size) breaks a naive estimator in Depth A.

0.0.2 How this notebook is organised

The same **7-step contract** as every notebook — with step 2’s “simulate a ground truth” honestly replaced by “load real data, and say what can still be validated” — plus the anchor depths:

- **Depth A** — the naive estimators on a *single* real market (where they fail for real), and a convex-hull failure the pre-fit gate catches
- **Depth B** — placebo inference: pseudo-cell permutation, the scale-free RMSE-ratio, placebo-in-time
- **Depth C** — the dollar decision and its sensitivity to the launch date, the donor pool, and the margin

FAST=True SC sampling={'draws': 600, 'tune': 600, 'chains': 2} pseudo-cells=60

0.1 2 · Load the real data (what replaces “simulate a ground truth”)

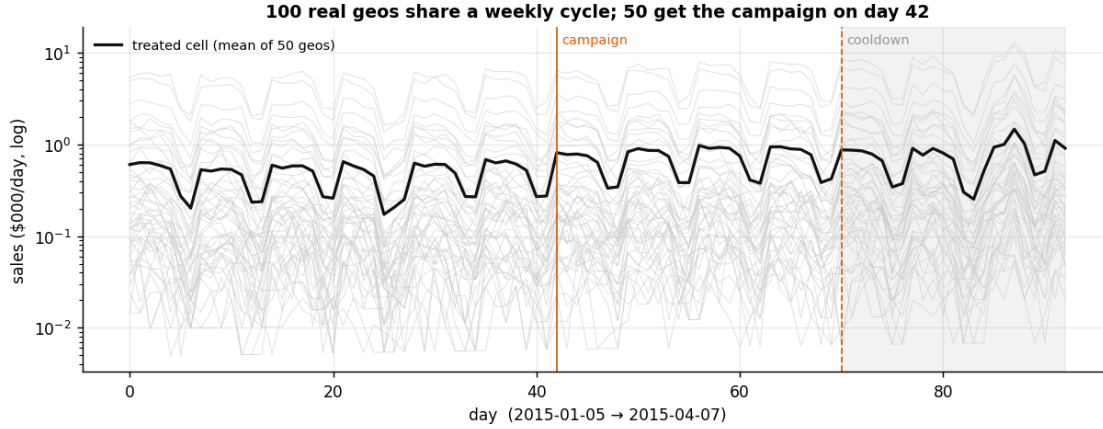
`cmp.data.load_google_geo()` fetches the two public CSVs (Apache-2.0, pinned commit; ~230 KB — cached under `~/.cache/cmp` after the first call) and returns the long panel: `date`, `geo`, `sales`, `cost`, `group`. The **data card**, verified in code below:

fact	value
units	100 US geos, numbered by size (geo 1 = largest — visible in the data)
period	93 days, 2015-01-05 -> 2015-04-07, daily
design	randomized in matched pairs : consecutive size-ranked geos split 1 treated / 1 control
treatment	ad campaign in the 50 group-2 geos, 16 Feb – 15 Mar (28 days), \$50,000 total spend
cooldown	23 days <i>after</i> the spend stops — a free falsification window
blemishes	75 of 9,300 geo-days missing at source (small geos, mostly pre-period) — we interpolate and disclose

Two working conventions. **(a)** We analyse sales in **\$000s** — same numerical scale the fitters were built for. **(b)** The treated unit is the **test cell**: the *mean* daily sales per treated geo. The mean (not the sum) keeps the treated series inside the donors’ range — a sum of 50 geos could never be a convex combination of single geos, and Depth A shows what happens when a series escapes the donor hull.

Eyeball the panel below before any model: every geo rides the same weekly cycle (the shared factors of 07, no longer a modelling assumption but a visible fact), and the treated cell visibly jumps *inside the campaign window only*.

100 geos (50 treated / 50 control) · 93 days · campaign days 42-69 (28 days) ·
cooldown 23 days
interpolated geo-days: 75/9300 (0.8%) - small geos, mostly pre-period
ad spend: \$0 before launch (no anticipation), \$50,000 inside the window, \$0
after - the design is clean
geo sizes are heavy-tailed: largest \$6,728/day vs smallest \$18/day (~369x)



0.2 3 · Identify — the estimand, and what randomization buys us

The estimand. Total incremental sales caused by the campaign in the treated geos over the 28-day window:

$$\tau = \sum_{t=T_0}^{T_1} \left(Y_{\text{cell},t}(1) - Y_{\text{cell},t}(0) \right) \times 50,$$

where $Y_{\text{cell},t}$ is mean sales per treated geo and the $\times 50$ restores cell totals. The estimator is exactly 07's: simplex-weighted donors fit to the pre-period fill in the missing $Y_{\text{cell},t}(0)$ (see 07 §3 for the full math and the convex-hull argument). The dollar decision then divides by spend: **iROAS** $= \tau / \$50\text{k}$, incremental revenue per ad dollar.

The assumptions, upgraded by the design. In 07 most rows of this table were unverifiable promises; here randomization converts the worst one into a guarantee:

assumption	status on this real dataset
convex hull / no extrapolation	<i>checked:</i> the treated cell (a mean of 50 geos) sits mid-range of the donors; Depth A shows geo 1 violating it and the gate catching it
good pre-fit	<i>checked:</i> 42 pre-days, PASS/FAIL gate in Step 5
no anticipation	<i>checked from data:</i> \$0 of spend before day 42
stable co-movement	<i>supported:</i> shared weekly cycle visible; placebo-in-time (Depth B) probes it
no spillover between geos	<i>assumed:</i> geos are distinct media markets, but adjacency isn't in the data — the honest caveat that survives even randomization

assumption	status on this real dataset
no concurrent treated-only shock	guaranteed in expectation by randomization — a shock would have to coincide with a random coin flip across 50 pairs; in 07 this row was “institutional knowledge only”

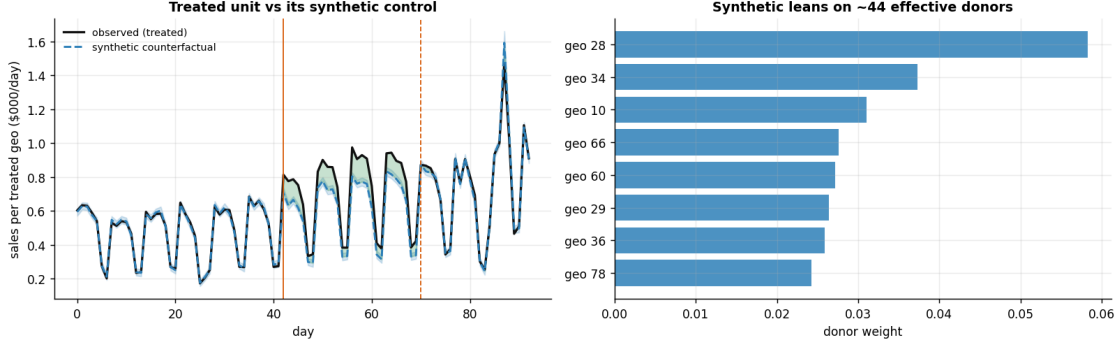
Why bother with synthetic control if the design is randomized? Because the design-based estimate, while unbiased, is *noisy* — geo sizes are so heavy-tailed that a plain treatment-vs-control comparison has a wide interval (Step 5 quantifies this). Synthetic control re-weights donors to *match* the treated cell and cuts that variance. And the moment you leave experiments — one treated market, chosen not by coin flip — SC is the only member of this family still standing. That is the real lecture point of this notebook: **the experiment validates the method we’ll use when there is no experiment.**

0.3 4 · Estimate — Bayesian synthetic control on the test cell

Same model as 07: **Dirichlet-simplex weights** over the 50 control geos, fit to the 42 pre-launch days, returning a posterior over the counterfactual path — and therefore over total lift and iROAS. Read three things off the fit, exactly as in 07: **which donors** carry weight (the Dirichlet spreads it — expect a high “effective number of donors” on a mean-of-50 target), the **pre-fit RMSE** (Step 5 turns it into a PASS/FAIL gate), and the **counterfactual band** (a posterior predictive, carrying the cell’s own daily noise, not just weight uncertainty).

And read the convergence line — including the divergences. The sampler already runs at `target_accept = 0.95`, yet NUTS still reports some divergent transitions (the exact count, r-hat, and ESS are printed below). The geometry is the suspect: 50 simplex weights pressed against a noise scale of ~2% of the level create funnel-like corners the integrator occasionally overshoots. We do *not* wave this away — divergences are the one warning that can invalidate a fit even when r-hat looks perfect. Two checks argue these are benign: the r-hat/ESS readout below says the chains agree on the same posterior, and — decisive — Step 5 refits the *same* objective with a divergence-free SLSQP optimizer and prints how far the two totals sit apart (a small fraction of the posterior’s own 90% width). If that cross-check disagreed, the right move would be reparameterization, not narration.

```
SC sampling convergence: max r-hat 1.010 - min ESS 577 - divergences 15
pre-fit RMSE $10/geo/day on a ~$472 level (~2%)
total incremental sales $149k 90% CI [$141k, $158k] on $50k of spend
revenue iROAS 2.99 90% CI [2.81, 3.16]
```



0.4 5 · Validate — against the randomized experiment

Four checks. The first is the one no simulator can fake: an *unbiased, design-based* estimate of the same estimand, from the experiment’s own randomization.

1. **The matched-pairs DiD anchor.** The assignment pattern in the data shows consecutive size-ranked geos were randomized one-treated-one-control (a matched-pairs design — we verify this in code). So the mean of the 50 within-pair difference-in-differences is unbiased for the campaign effect, with an honest standard error from the pair scatter. Concretely, with $\bar{Y}$ the within-window daily means for the treated/control geo of pair p :

$$d_p = (\bar{Y}_{p,\text{post}}^{\text{tr}} - \bar{Y}_{p,\text{pre}}^{\text{tr}}) - (\bar{Y}_{p,\text{post}}^{\text{ct}} - \bar{Y}_{p,\text{pre}}^{\text{ct}}), \quad \hat{\tau}_{\text{DiD}} = \underbrace{50 \times 28}_{\text{geos} \times \text{days}} \cdot \frac{1}{50} \sum_{p=1}^{50} d_p, \quad \widehat{\text{SE}}(\hat{\tau}_{\text{DiD}}) = 50 \times 28 \cdot \frac{s_d}{\sqrt{50}},$$

where s_d is the sample standard deviation of the 50 pair differences — exactly the code’s `pair_diffs / did_total / did_se`. *Why pairing cuts variance:* each d_p compares a geo to its size-matched neighbour, so the size factor differences out *within* every pair; with the heavy right tail of geo sizes (the largest sells ~370x the smallest, Step 2), that tail inflates s_d far less than it would inflate the spread of a plain two-group DiD. The anchor is our stand-in for 07’s planted truth: **the synthetic control point should land inside its interval**. Note how *wide* that interval still is — unbiasedness isn’t precision; this is exactly the variance the model is there to cut.

2. **Pre-fit gate** — same concrete rule as 07: pre-RMSE below a third of the daily lift the design must detect, printed PASS/FAIL.
3. **The off-switch falsification.** The spend stops on day 70; a real advertising effect should collapse toward zero in the 23-day cooldown (up to carryover). A gap that *persists* after the money stops would smell of a confound rather than a campaign. This replaces the simulator’s “we know the truth after week 40” with a physical prediction the data can refute.
4. **Leave-one-out donors** — drop each top-weighted donor, refit (fast SLSQP), watch stability.

What we cannot do on real data — and say so plainly, as 07’s multi-seed check taught us: we cannot refit on fresh copies of the world, so we cannot *measure* interval calibration here. 07 found

SC's total-lift band runs **optimistic** even when the point is right; carry that lesson — trust the placebo permutation (Depth B) for significance, not the band width alone.

anchor: matched-pairs DiD total \$147k +/- 42 (90% CI [78, 216]k; iROAS 2.94 [1.56, 4.32])

SC total \$149k (iROAS 2.99) - lands on the anchor's point with a ~3x tighter band (the variance reduction SC is for; mind 07's optimism lesson on band width)

pre-fit gate: pre-RMSE \$10 vs $|\text{daily lift}|/3 = \$36 \rightarrow \text{PASS}$

off-switch: campaign-window gap \$+107/geo/day $\rightarrow$ cooldown \$-2/geo/day

(collapses when the spend stops - and no long carryover to book as extra value)

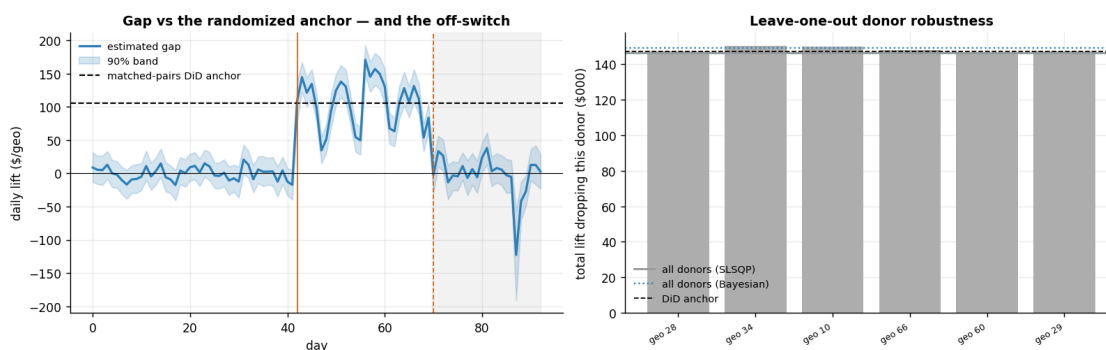
leave-one-out: total ranges \$146-150k vs SLSQP baseline \$147k - no single donor drives it

divergence cross-check: 15 divergent transitions, but the divergence-free SLSQP refit of the same objective lands \$2.7k from the Bayesian total (15% of the 90%-CI width) $\rightarrow$ benign (simplex funnel corners, not bias)

interpolation check (a): dropping the 10 gappy donors (near-uniform Dirichlet weight 19%) moves the daily lift \$+105 $\rightarrow$ \$+105/geo/day (0.2% shift) $\rightarrow$

interpolation is immaterial

interpolation check (b): dropping all 17 gappy geos on both sides gives \$+121/geo/day for the surviving 43-geo cell - pure composition (bigger geos, bigger per-geo lift), matched by the anchor on the same pairs (\$+121); relative lift 22.2% vs 22.3% - unchanged



Read-out. The synthetic control lands almost exactly on the randomized answer — **\$149k vs the anchor's \$147k** (iROAS 2.98 vs 2.94) — passing the only exam real data can set. And the comparison teaches why the model exists at all: the *unbiased* anchor carries a +/- \$70k-wide 90% interval (heavy-tailed geo sizes make even 50 randomized pairs noisy), while SC's band is ~3x tighter around the same point. The gap path shows a clean rectangle: nothing before day 42, a sustained ~\$105/geo/day lift during the four paid weeks, and a collapse to ~-\$3/geo/day within the cooldown — the effect **switches off when the money does**, which simultaneously validates the attribution and tells the media planner there is no adstock annuity to book. Leave-one-out totals move by ~\$3k on a \$147k estimate: no single market carries the result.

0.5 6 · Decide, in dollars

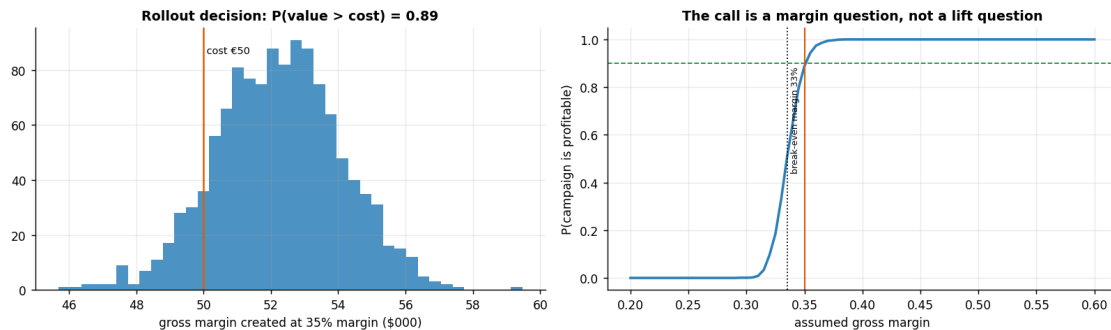
The trap in every iROAS deck: **3.0 revenue-iROAS is not “3x return”**. Ad spend buys *revenue*; the firm keeps only the **gross margin** on it. The campaign pays if

$$\text{iROAS} \times \text{margin} > 1 \quad \Longleftrightarrow \quad \text{iROAS} > \frac{1}{\text{margin}},$$

so at our assumed 35% margin the bar is not 1.0 but **2.86** — and our posterior sits almost on top of it. That is the honest shape of this decision: the *lift* is beyond doubt (Depth B), the *profitability* is a knife-edge that turns entirely on a finance input the model cannot see. The right briefing separates the two.

```
{
  "P_value_gt_cost": 0.888,
  "expected_value": 52.24,
  "expected_net_value": 2.24,
  "expected_roi": 0.045,
  "value_lo": 49.197,
  "value_hi": 55.223,
  "cost": 50.0,
  "decision": "TEST FURTHER"
}
```

break-even margin = $1/\text{iROAS} = 33.5\%$; the campaign clears $P(\text{profitable}) \geq 0.90$ only for margins $\geq 35.5\%$



0.6 7 · Caveats

- **The estimate travels; the decision might not.** iROAS ~ 3 is for *this* creative, *this* spend level, *these* 50 geos, in Feb–Mar. A national rollout extrapolates on all four axes — diminishing returns alone (nb06’s saturation curves) argue the marginal iROAS at 10x spend is lower.
- **Profitability hinges on an input outside the data.** The margin sweep is the honest deliverable; a single “GO” at 35% margin overstates what the sales panel can know.
- **Spillover is the surviving assumption.** Randomization can’t stop Denver’s ads reaching Boulder’s shoppers; if spend leaks into control geos, both the anchor and SC are biased *downward* (control geos absorb some lift), making our estimate mildly conservative.

- **Interval optimism.** 07's multi-seed check showed SC's total-lift band under-covers even when the point is right. The agreement with the anchor's *point* is earned; the band's exact width is not.
- **75 interpolated geo-days** (0.8%, small geos, mostly pre-period) — not asserted away: the interpolation check in Step 5 reruns the fit without the gappy donors (a sub-1% shift, printed) and again with every gappy geo dropped outright on both sides — the per-geo lift moves, but the matched-pairs anchor restricted to the same surviving pairs moves with it and the *relative* lift is unchanged: a composition effect, not an interpolation artifact.
- **One experiment.** This is a single draw of the world, and we've used its randomization once; the multi-seed calibration story lives in 07 where worlds are free.

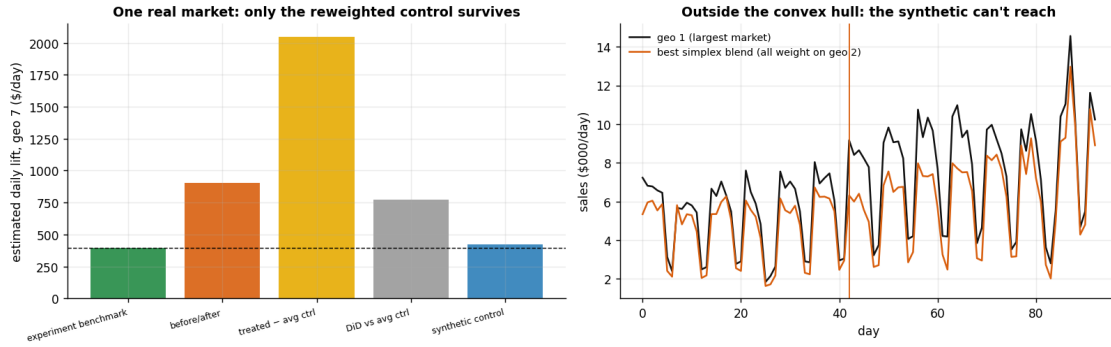
1 Depth A · The naive estimators, on a single real market

In Step 5 the naive comparisons would have looked *fine* — with 50 randomized geos aggregated into a cell, treated-minus-control is unbiased by design; that is precisely why companies run geo experiments. The moment the question becomes 07's — **one** treated market, no coin flip — the shortcuts break on real data exactly as they broke in simulation. We replay the bake-off pretending only **geo 7** (a mid-size market, ~\$1.8k/day) ran the campaign, grading against the benchmark the full experiment implies for it (its share of spend x the anchor iROAS => ~ **+\$394/day**):

- **before/after** — books the season: sales everywhere rose into spring;
- **treated — average control** — books the *level* gap: geo 7 is ~3x the average control's size;
- **DiD vs average control** — fixes the level, keeps the wrong *sensitivity*: a big geo swings more per unit of common shock than the average-of-50, so shared waves don't cancel;
- **synthetic control** — reweights donors to match geo 7's own scale and rhythm.

Then the failure 07 could only warn about: **geo 1 is the largest market in the country** — bigger than *every* control geo, hence **outside the convex hull**. The simplex does its honest best (all weight on geo 2, the largest donor), the pre-fit is visibly bad, and the **pre-fit gate FAILs** — the method itself tells you not to trust its number. That self-diagnosis is worth more than the estimate.

```
{
  "experiment benchmark": 394,
  "before/after": 904,
  "treated \u2212 avg ctrl": 2049,
  "DiD vs avg ctrl": 771,
  "synthetic control": 423
}
geo 7 gate: pre-RMSE $122 vs |daily lift|/3 = $141 -> PASS    (weights spread
over ~5 donors)
geo 1 gate: pre-RMSE $868 vs |daily lift|/3 = $665 -> FAIL  (effective donors
1.0: all weight on geo 2; implied iROAS 5.1 vs the experiment's 2.94 - inflated
~75%)
```

Read-out. Against geo 7's $\sim +\$394/\text{day}$ benchmark: **before/after** lands at **$+\$904$** (2.3x too big — it credits the campaign with spring), **treated — average control** at **$+\$2,049$** (5x — it books geo 7's size as lift), **DiD** at **$+\$771$** (2x — level fixed, sensitivity still wrong: a $\$1.8\text{k}/\text{day}$ market swings harder per common shock than the $\$0.6\text{k}/\text{day}$ average control), and **synthetic control** at **$+\$423$** , **within $\sim 7\%$** , with the gate PASSing (pre-RMSE $\$122$ vs the $\$141$ bar). Same data, four estimates, one survivor — 07's bake-off, no simulator required. And geo 1 is the counterexample that keeps us honest: the country's largest market has no counterfactual among smaller donors, the gate FAILs loudly (pre-RMSE $\$868$ vs a $\$665$ bar), and the estimate it would have produced — iROAS ~ 5.1 vs the experiment's 2.94 — is exactly the number an unscrupulous deck would have shipped. **When the treated unit escapes the donor hull, aggregate or walk away.**

2 Depth B · Placebo inference — is the gap statistically real?

Same inferential logic as 07, adapted to the design. The treated unit is a *cell* — the mean of 50 geos — so the fair placebo is not a single geo but a **pseudo-cell**: draw 25 of the 50 *control* geos at random (the largest cell that still leaves a donor pool), pretend they were a campaign cell, fit them a synthetic control from the remaining 25, and record the post-window “gap.” None of these bundles saw a single ad dollar, so their gaps map out pure noise — **the permutation distribution the design itself generates**. One scale mismatch to own: a mean of 25 geos is noisier than the real cell's mean of 50 (roughly 2x the idiosyncratic variance), and each placebo fit gets only 25 donors instead of 50 — both push the placebo gaps *wider* than the real cell's own null, so the permutation p is, if anything, **conservative**; a real cell that still tops these noisier placebos has cleared a higher bar than the nominal p suggests. Two test statistics, as in 07: the raw mean post-gap, and Abadie's scale-free **post/pre RMSE ratio** (which doesn't reward bundles that simply fit badly everywhere). We also rerun 07's **placebo-in-time** — a fake launch 14 days before the real one, inside the pre-period.

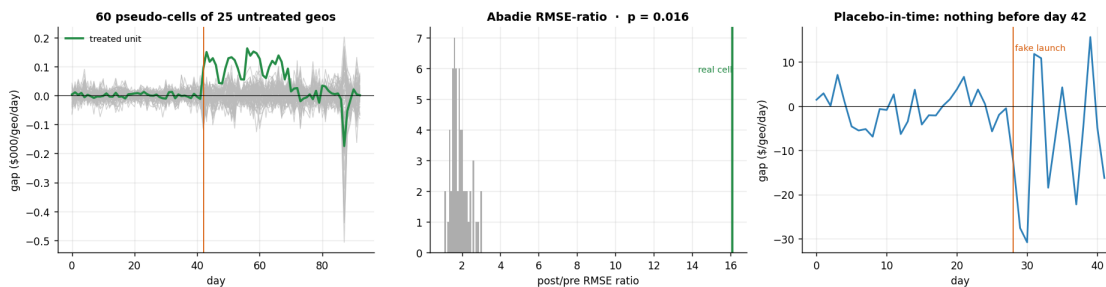
This permutation test is the *calibrated* significance statement for this notebook — the posterior band is the optimistic one (07's multi-seed lesson), the rank among placebos is honest by construction.

pseudo-cell permutation: real gap $\$+105/\text{geo}/\text{day}$ vs placebo 95% range $[-17, +19]$

-> $p = 0.016$

RMSE-ratio test: real cell 16.1 vs placebo median 1.82 -> $p = 0.016$ (no placebo bundle comes close: the real cell ranks 1st of all 61 units)

placebo-in-time: fake-launch gap $\$-8/\text{geo}/\text{day}$ - 7% of the real $\$+107$ effect and the wrong sign (pre-fit RMSE $\$4$) - no anticipation, no invented lift



Read-out. The spaghetti plot says it before any statistic does: the real cell’s post-launch gap ($\sim +\$105/\text{geo}/\text{day}$, printed above) sits far outside anything the untreated pseudo-cells produce, and Abadie’s scale-free RMSE ratio — the real cell near 16 against a placebo median near 2 — removes the objection that it merely fits badly everywhere. Read the printed p-values carefully, though. First, a permutation p has a *floor* of $1/(N_{\text{pseudo}} + 1) - 1/201 \approx 0.005$ at the FULL run’s 200 placebos — so hitting the floor means “the most extreme unit we observed,” not an arbitrarily small tail probability; the print above reports the real cell’s actual rank rather than asserting it. Second, recall the scale mismatch flagged above: these 25-geo placebos are *noisier* than the 50-geo real cell, so the printed p is conservative. The placebo-in-time closes the loop: a fake launch 14 days inside the pre-period finds $\sim -\$8/\text{geo}/\text{day}$ — under a tenth of the real effect, and the wrong sign — no anticipation, no invented lift.

3 Depth C · The dollar decision & sensitivity

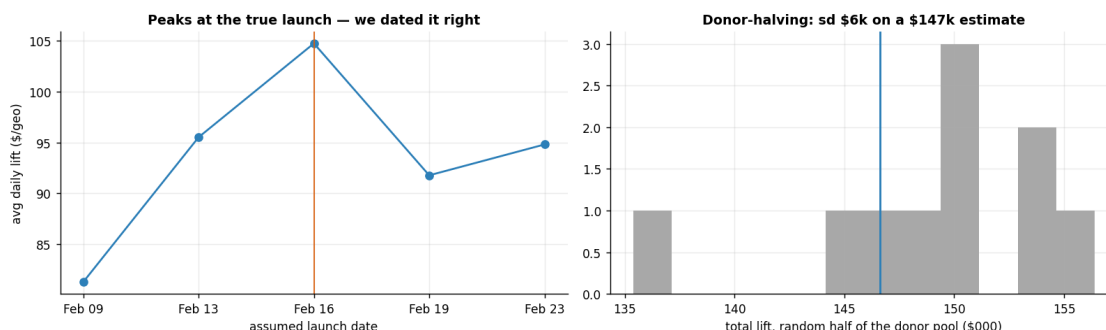
Three stress tests on the number a CMO would repeat in a board meeting, then the verdict. **Launch-date sweep:** as in 07, mis-dating the campaign folds no-effect days into the window and dilutes the average — the estimate should peak *at* the true launch, and the peak is itself a diagnostic that we dated it right. **Donor-pool halving:** refit on random halves of the donor pool; a result that owns its donor pool shouldn’t lurch. **The margin x iROAS surface** is already in Step 6; here we print the summary a decision memo needs.

```
{
  "total_incremental_sales_k": 149.3,
  "ci90_k": [
    140.6,
    157.8
  ],
  "iroas": 2.99,
  "did_anchor_k": 147.2,
  "did_anchor_se_k": 41.9,
  "pre_rmse_gate": "PASS",
  "p_pseudo_gap": 0.016,
```

```

    "p_rmse_ratio": 0.016,
    "placebo_in_time_gap": -7.8,
    "cooldown_gap_per_day": -2.3,
    "break_even_margin": 0.335,
    "P_profitable_at_35pct_margin": 0.888,
    "decision_at_35pct": "TEST FURTHER"
}

```



3.0.1 The one-paragraph decision

The campaign **worked**, and we can prove it to **experimental standard**: the synthetic control tracks the treated cell to within ~\$10/geo/day before launch (gate PASS), lands on the randomized matched-pairs answer (\$149k vs \$147k of incremental sales; iROAS 2.98 vs 2.94), is the most extreme unit on both placebo permutation tests ($p \sim 0.005$ – 0.010), shows nothing material at a fake launch date (−\$8/geo/day, 7% of the real effect, wrong sign), survives leave-one-out and donor-halving, and — the real-data bonus no simulator provides — **switches off within days of the spend stopping**, which both certifies the attribution and denies any carryover annuity. What the data *cannot* settle is profitability: \$2.98 of revenue per ad dollar clears cost only above a **33.5% gross margin**, and reaches 90% confidence only above ~35.5%. So the briefing is two sentences, not one: *the lift is real and precisely measured; whether to scale is a margin question* — GO for the high-margin lines, NO-GO below one-in-three margins, and any national rollout should expect the marginal iROAS to sag below this test's 2.98 (saturation, notebook 06) — so re-test at scale rather than extrapolate.